Windows Networking Services

- **File Server Role Theory :File server overview and use cases. Configuring shared folders with permissions. Practical :Set up shared folders and assign NTFS and share permissions. Task: Test file access from client systems and document results.**

1. On your Windows Server VM, create a shared folder named 'MusicLibrary' and set the share permissions so only the 'Students' group has read access.

ANS:

Step 1: Create the folder

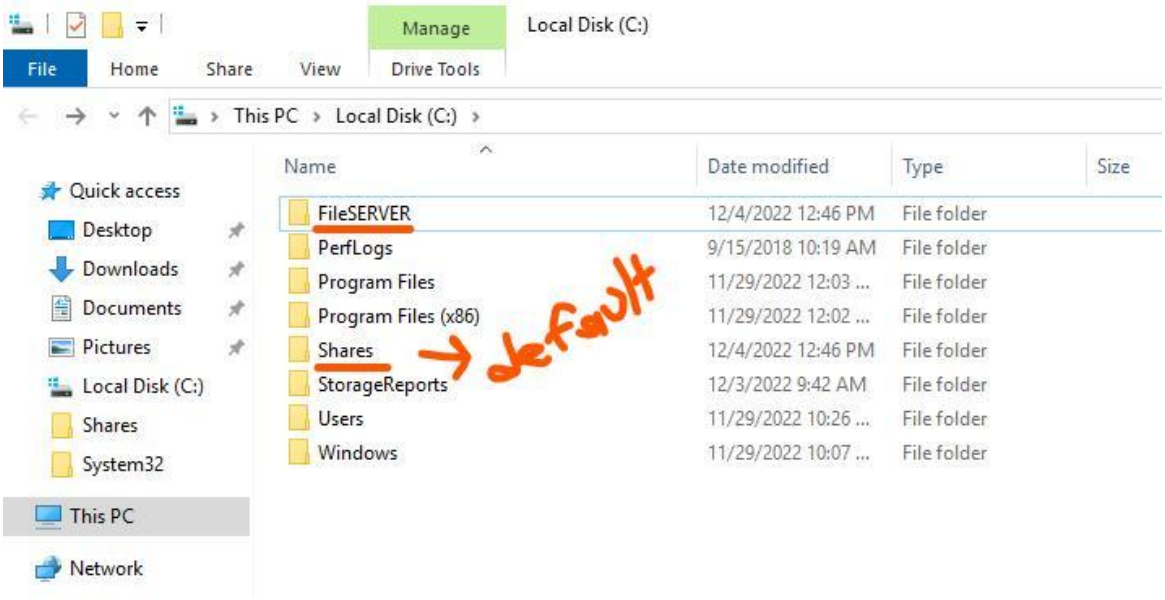
1. Log in to your Windows Server.
2. Open File Explorer.
3. Navigate to the C:\ drive.
4. Create a new folder named MusicLibrary.

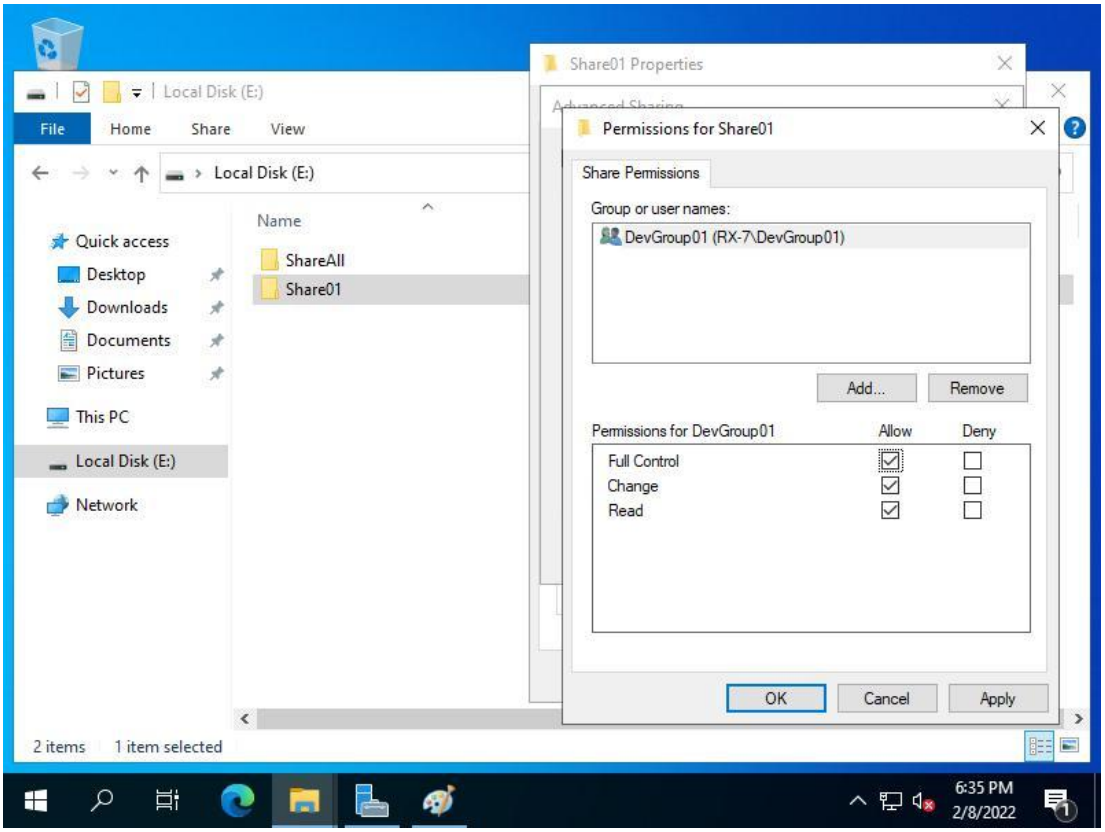
Step 2: Open folder properties

1. Right-click the MusicLibrary folder.
2. Select Properties.
3. Open the Sharing tab.
4. Click Advanced Sharing.
5. Check Share this folder.
6. Share name: MusicLibrary.
7. Click Permissions.

Step 3: Configure Share Permissions

1. Remove the Everyone group.
2. Click Add.
3. Enter Students.
4. Click Check Names.
5. Click OK.
6. Select the Students group.
7. Allow only Read permission.
8. Click Apply and OK.


Manage
Local Disk (C:)
File
Home
Share
View
Drive Tools
This PC › Local Disk (C:) ›
Quick access
Desktop
Downloads
Documents
Pictures
Local Disk (C:)
Shares
System32
This PC
Network
Name
Date modified
Type
Size
FileSERVER 12/4/2022 12:46 PM File folder
PerfLogs 9/15/2018 10:19 AM File folder
Program Files 11/29/2022 12:03 ... File folder
Program Files (x86) 11/29/2022 12:02 ... File folder
Shares 12/4/2022 12:46 PM File folder
StorageReports 12/3/2022 9:42 AM File folder
Users 11/29/2022 10:26 ... File folder
Windows 11/29/2022 10:07 ... File folder
default


Local Disk (E:)
File
Home
Share
View
Local Disk (E:)
Quick access
Desktop
Downloads
Documents
Pictures
This PC
Local Disk (E:)
Network
Name
ShareAll
Share01
2 items
1 item selected
Share01 Properties
Permissions for Share01
Share Permissions
Group or user names:
DevGroup01 (RX-7\DevGroup01)
Add...
Remove
Permissions for DevGroup01
Allow
Deny
Full Control
Change
Read
OK
Cancel
Apply
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**2. Configure NTFS permissions on the 'MusicLibrary' folder so that members of the 'Students' group can read and write files, but cannot delete any files.

Hint: Use the Advanced Security Settings to deny delete permissions while allowing modify.**

ANS:

Step 1: Open Advanced Security Settings

1. Right-click MusicLibrary.
2. Select Properties.
3. Open the Security tab.
4. Click Advanced.

Step 2: Add Students Group

1. Click Add.
2. Select Select a principal.
3. Enter Students.
4. Click Check Names and OK.
5. Set the permission type to Allow.
6. Select Show advanced permissions.

Configure the following permissions:

Advanced Permission	**Setting**
Traverse folder / Execute file	Allow
List folder / Read data	Allow
Read attributes	Allow
Read extended attributes	Allow
Create files / Write data	Allow
Create folders / Append data	Allow
Write attributes	Allow
Write extended attributes	Allow
Read permissions	Allow
Delete	Deny
Delete subfolders and files	Deny

Step 3: Apply Permissions

1. Set Applies to as This folder, subfolders and files.
2. Click Apply.
3. Click OK.
4. Confirm the permission changes.

3. From a Windows client system, map the 'MusicLibrary' shared folder as a network drive and try to copy, edit, and delete a file. Document which actions succeed or fail, and explain why based on the permissions you set.

ANS:

Step 1: Find the Server IP Address

On your Windows Server, open Command Prompt and execute:

ipconfig

Note the IPv4 address of your server.

Example: 192.168.1.119

Step 2: Access the Shared Folder

On your Windows client:

1. Press Win + R.
2. Enter the following command: \\192.168.1.119\MusicLibrary
3. Press Enter.
4. If prompted, enter the credentials of a user who belongs to the Students group.

Replace the IP address with your actual server IP.

Step 3: Map Network Drive

1. Open File Explorer.
2. Right-click This PC.
3. Select Map network drive.
4. Select a drive letter, such as Z:.
5. Enter : \\192.168.1.119\MusicLibrary
6. Check Reconnect at sign-in if required.
7. Click Finish.

Step 4: Test File Permissions

Perform the following operations using a Students group account.

Operation	Expected Result	Reason
Open folder	Success	Share and NTFS Read allowed
Read file	Success	Read permission allowed
Copy file from share to client	Success	Read permission allowed
Create a new file	Success	NTFS Write allowed
Edit an existing file	Success	NTFS Write allowed
Delete a file	Failed	NTFS Delete explicitly denied

4.Create a second shared folder called 'PhotoShare' and set up share and NTFS permissions so that only your user account has full control. Attempt to access this share from another user account and note the result.

ANS:

Step 1: Create PhotoShare Folder

1. Open File Explorer.
2. Navigate to C:\.
3. Create a folder named: PhotoShare

Step 2: Configure Share Permissions

1. Right-click PhotoShare.
2. Select Properties.
3. Open the Sharing tab.
4. Click Advanced Sharing.
5. Check Share this folder.
6. Set Share name to PhotoShare.
7. Click Permissions.
8. Remove Everyone.
9. Add your user account.
   For example: Administrator
10. Select your user account.
11. Allow Full Control.
12. Click Apply and OK.

Step 3: Configure NTFS Permissions

1. Open the Security tab.
2. Click Advanced.
3. Remove unnecessary user/group entries that grant other users access.
4. Add your own user account.
5. Select the following permissions:

Permission	Setting
Full Control	Allow
Modify	Allow

Read & Execute	Allow
List Folder Contents	Allow
Read	Allow
Write	Allow

6. Set Applies to: This folder, subfolders and files.
7. Click Apply and OK.

Step 4: Test Access from Another User Account

1. Log in to your Windows client using another user account.
2. Press Win + R.
3. Enter : \\192.168.1.119\PhotoShare
4. Press Enter.

